How Policy and Regulatory Uncertainty Shapes Renewable Energy Consumption in the U.S.-Mexico Border States: A Time Series Analysis, January 2001–April 2026

The main series, Renewable Energy Consumption (REC), is monthly net electricity generation from renewable sources, all sectors, in thousand megawatthours (MWh), for Texas, New Mexico, Arizona, and California. REC is the sum of conventional hydroelectric, wind, utility-scale solar, geothermal, and biomass generation. I follow Tao et al. (2025) in calling the series renewable energy consumption, although the U.S. Energy Information Administration (EIA) reports generation, not end-use consumption. The data come from EIA's Electric Power Monthly, accessed through the EIA Open Data API v2 (https://www.eia.gov/opendata/) on October 2, 2026.

A second series excluding hydroelectric generation is kept as a robustness check. It is exactly zero in eight early months in New Mexico and nearly zero in Arizona, so it cannot be log-transformed in those states. Every file is downloaded unmodified by a script in the project repository (code/01_download_and_clean.py), which records the access date of each source. All variables are listed in Table 1.

Table 1. Variables and sources

Variable	Definition and unit	Source and access	Frequency
REC	Net generation, all renewables incl. hydro; thousand MWh	EIA Electric Power Monthly via EIA API v2 (https://www.eia.gov/opendata/)	Monthly, by state
REC excl. hydro	Same, without conventional hydro (robustness)	Same	Monthly, by state
EPU	News-based economic policy uncertainty index	Baker, Bloom, and Davis (2016); https://www.policyuncertainty.com	Monthly, national
CPU	News-based climate policy uncertainty index (narrow dictionary)	Gavriilidis et al. (2026); https://github.com/dkaenzig/Climate-Policy-Uncertainty-Index	Monthly, national
GPR	Geopolitical risk index (robustness)	Caldara and Iacoviello (2022); https://www.matteoiacoviello.com/gpr.htm	Monthly, national
WTI	Crude oil spot price; USD per barrel	EIA API v2, series PET.RWTC.M	Monthly, national
Henry Hub	Natural gas spot price; USD per MMBtu	EIA API v2, series NG.RNGWHHD.M	Monthly, national
Retail price	Average retail electricity price, all sectors; cents per kWh (EIA computes it from revenue and sales); deflated by the CPI	EIA Form EIA-861M, API v2 route electricity/retail-sales; CPI: FRED series CPIAUCSL [pending]	Monthly, by state
State activity	State coincident economic index [pending]	Federal Reserve Bank of Philadelphia, via FRED (TXPHCI, NMPHCI, AZPHCI, CAPHCI)	Monthly, by state
Regulatory events	Dated indicators of state RPS enactments or amendments and restructuring suspensions [pending]	DSIRE (https://programs.dsireusa.org) and EIA	Event dates (dummy variables)

Sample period and frequency. The sample runs from January 2001 to April 2026, 304 monthly observations per state. January 2001 is the earliest month for which EIA publishes monthly state generation

by fuel type, and April 2026 is the last month of the climate policy uncertainty index, which limits the common sample.

Monthly frequency is chosen for three reasons. First, the events of interest (the 2008 financial crisis, COVID-19 in March 2020, Winter Storm Uri in February 2021, and state regulatory changes) occurred in specific months, so structural break tests can locate them much more precisely than with annual data. Second, 304 observations per state support the VAR and regime-switching models used in the course. Third, every explanatory series is published monthly, so no variable requires interpolation; variables available only annually, such as population, are excluded for that reason. Events before 2001, such as the California energy crisis and the 2002 restructuring suspensions, fall outside the testable window and are treated as historical context.

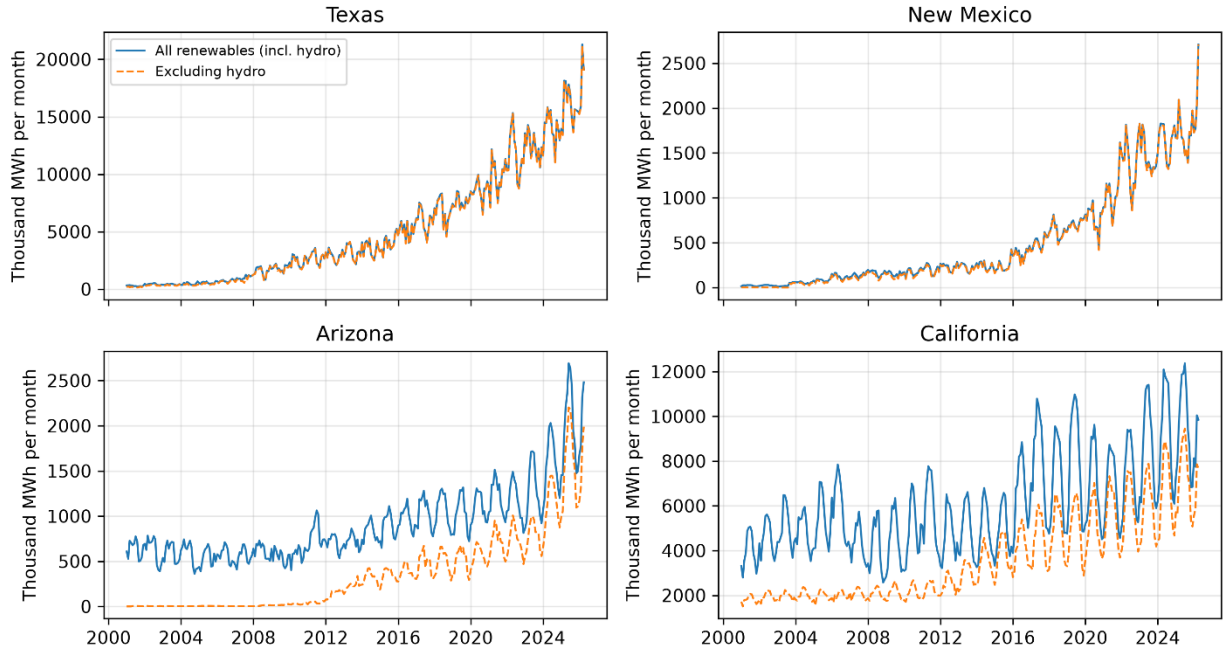
Transformations. Figure 1 shows exponential growth in generation and seasonal swings whose amplitude grows with the level, so REC, the policy indices, and the price series are expressed in natural logarithms [confirm no zeros in each series]. Renewable generation is strongly seasonal, so I will apply seasonal differencing at lag 12 / include monthly dummies [decide after the unit-root tests]. The order of differencing will be selected with ADF and KPSS tests. Retail prices are deflated by the CPI to remove inflation over 25 years. The regulatory indicators enter as dummy variables without transformation.

Summary statistics and figure. Table 2 reports the raw series. Generation in Texas rises from a minimum of 217.8 to a maximum of 21,299.0 thousand MWh per month, and its mean (5,115.7) is well above its median (3,240.4), reflecting a strong upward trend. Figure 1 shows pronounced seasonality in all four states and a composition shift in Arizona, where hydroelectric generation dominates until about 2011.

Table 2. Summary statistics, January 2001 to April 2026

Variable	State	N	Mean	Median	Std. dev.	Min	Max
REC (incl. hydro)	Texas	304	5,115.7	3,240.4	4,879.5	217.8	21,299.0
REC (incl. hydro)	New Mexico	304	514.0	233.9	565.7	11.4	2,709.8
REC (incl. hydro)	Arizona	304	904.7	770.7	417.9	361.4	2,691.2
REC (incl. hydro)	California	304	6,065.8	5,510.2	2,221.7	2,572.7	12,371.1
Other variables			[rows from table1_all_variables.csv]				

Figure 1. Monthly renewable net electricity generation by state (raw series)



Source: U.S. Energy Information Administration, Electric Power Monthly (API v2), 2001-01 to 2026-04. Vertical scales differ across panels.

References.

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